

Auteur: Sabawoon Enayat
Studierichting: Informatica
Oefeningenreeks 4A nr. 8.11

$$\begin{aligned}\text{Opgave: } & \int \frac{dx}{\tan(3x+5)} \\ \Rightarrow t = 3x+5, t' = 3, dt = t'dx \Rightarrow dx &= \frac{dt}{3} \\ = \int \frac{dt}{3 \tan t} = \frac{1}{3} \int \frac{1}{\frac{\sin t}{\cos t}} dt &= \frac{1}{3} \int \frac{\cos t}{\sin t} dt \\ \Rightarrow u = \sin(t), u' = \cos(t), du = u'dt = \cos(t)dt \Rightarrow dt &= \frac{dt}{u'} \\ \Rightarrow \frac{\cos t}{\sin t} \frac{1}{\cos t} = \frac{1}{\sin t} = \frac{1}{u} \\ = \frac{1}{3} \int \frac{1}{u} du = \frac{1}{3} \ln(|u|) + c \\ = \frac{1}{3} \ln(|\sin t|) + c = \frac{1}{3} \ln(|\sin(3x+5)|) + c\end{aligned}$$

References